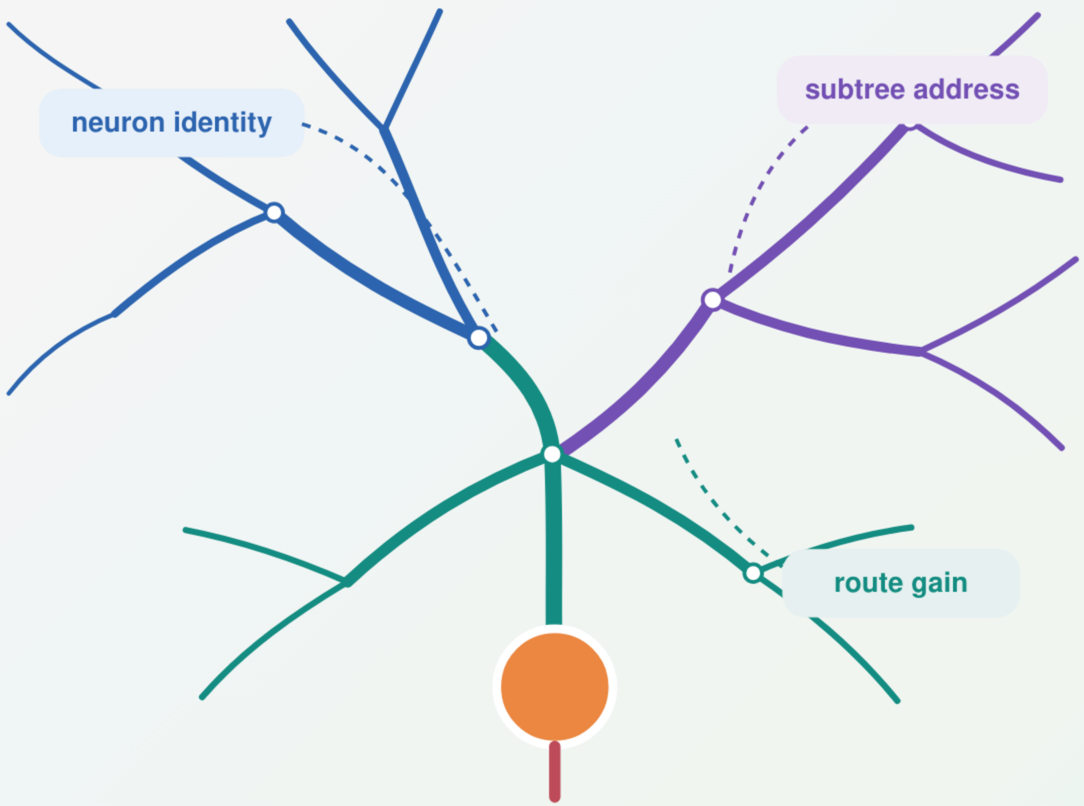
When can dendritic structure help local credit assignment?

Feedback bandwidth, within-neuron address, conductance-dependent route gain, and an alignment boundary

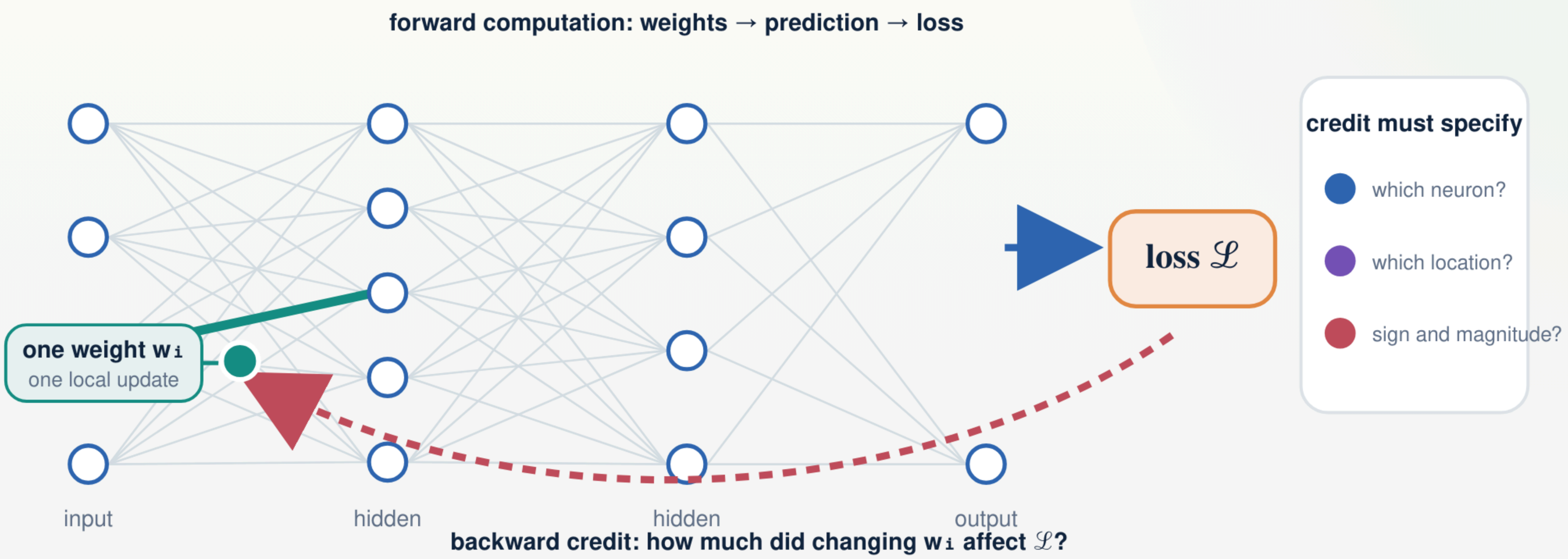
Houman Safaai · Maceo Richards · Bernardo L. Sabatini

Kempner Institute, Harvard University · Harvard Medical School



THE PROBLEM

Learning changes network weights; credit assigns each change

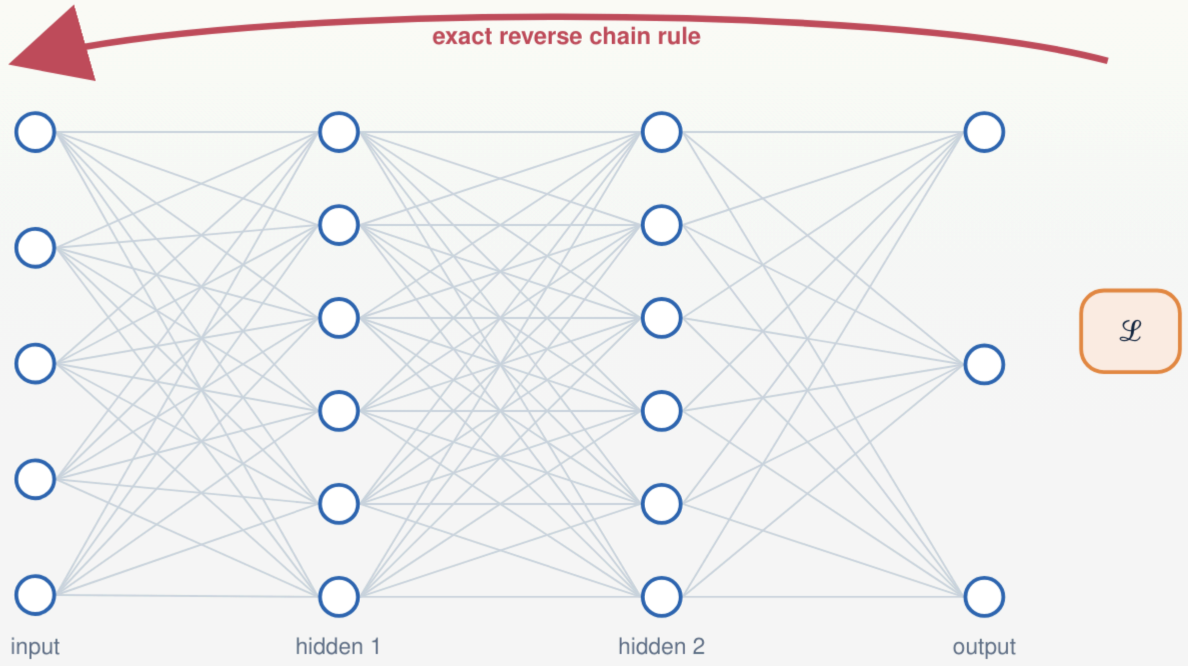


$$\mathcal{L} = \mathcal{L}(f(x; w), y) \quad \cdot \quad \Delta w_i = -\eta \, \partial\mathcal{L}/\partial w_i$$

→ **A global objective must be converted into a destination, sign, and magnitude for every trainable weight.**

THE MATHEMATICAL REFERENCE

Backpropagation defines the exact neuron-specific learning signal



$$z_u = \Sigma_{i \in in(u)} w_i x_i + b_u, \quad y_u = f_u(z_u)$$

$$\delta_u \equiv \partial\mathcal{L}/\partial y_u = \Sigma_{v:u\to v} w_{uv} f'_v(z_v)\delta_v$$

δ_u is exact task information assigned to neuron **u**—not yet to one weight.

fixed/random feedback	feedback alignment; direct feedback
inferred targets or states	equilibrium propagation; predictive coding
eligibility + learning signal	three-factor rules; e-prop
dendritic teaching signals	somato-dendritic and apical-error models

→ **Backpropagation is the information reference; biological theories differ in how the returned signal is produced.**

Rumelhart et al. (1986); Lillicrap et al. (2016); Scellier & Bengio (2017); Bellec et al. (2020)

Local learning preserves eligibility and approximates the returned signal

$$\partial\mathcal{L}/\partial w_i = \boxed{x_i\, f'_u(z_u)} \times \boxed{\delta_u}$$

local eligibility e_i

exact neuron signal

$$\Delta w_i = -\eta\, e_i\, \delta^{avail}_u$$

one layer-wide scalar m

Every neuron receives the same task-dependent coordinate.

one signal δ^{avail}_u per neuron

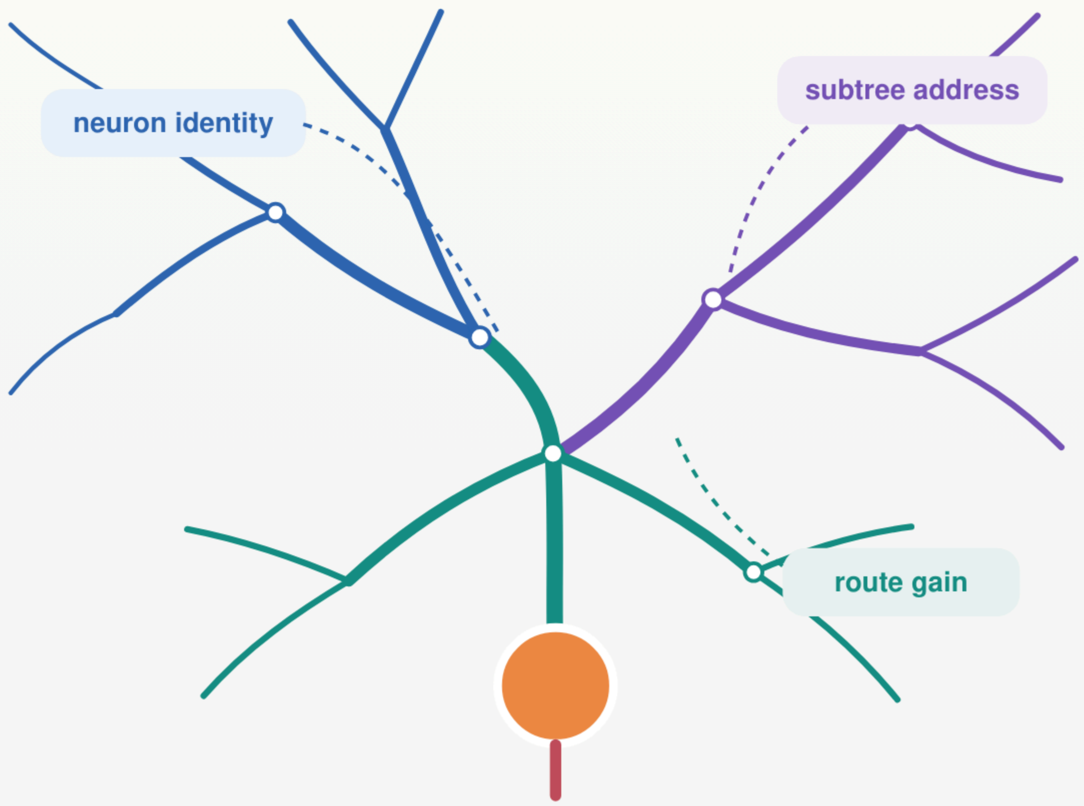
Feedback identifies the postsynaptic neuron while the eligibility remains weight-specific.

A shared scalar does not equalize the weights: each update still contains a different eligibility e_i.

→ **Locality determines where an update is computed; feedback bandwidth determines what task information it can use.**

Werfel, Xie & Seung (2005); Frémaux & Gerstner (2016)

Dendrites add within-neuron state, address, and route gain



network weight w_i → synaptic conductance g_i on compartment n

$$\delta^{V,\,\mathrm{avail}}_u = A_u\beta_u, \quad A_u \in \mathbb{R}^{N_u \times K}$$

K: independently communicated within-neuron signals · column k of $\mathbf{A}_u$: support and gain of route k

local state

Voltage and conductance determine the eligibility at each synapse.

subtree address

A few coefficients β_u can target nested groups of synapses.

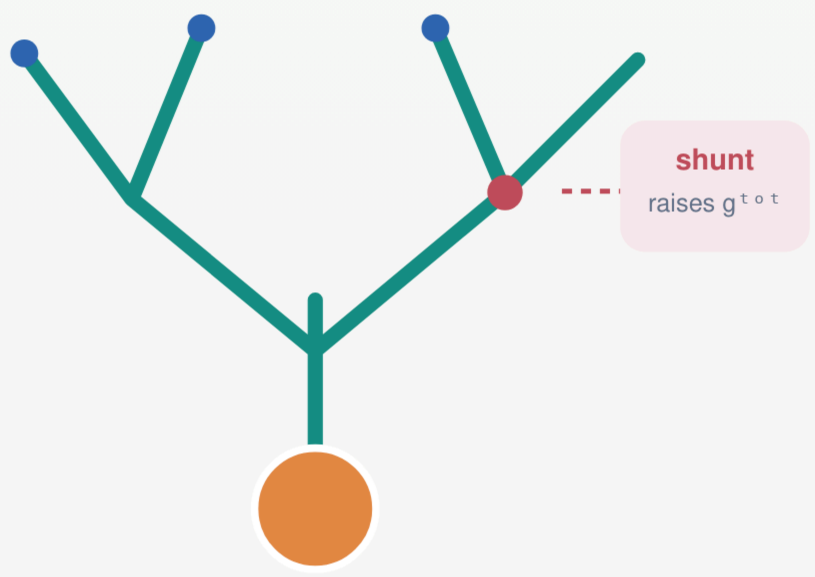
route gain

Conductance can scale how strongly a returned signal reaches one path.

→ **Neuron identity and within-neuron address are distinct credit-assignment problems.**

DIRECTED-TREE STEADY STATE

Conductance makes dendritic voltage a normalized quotient



$$g^{tot}_n = g^L_n + \Sigma_{i\in syn(n)} g_i x_i + \Sigma_{c\in child(n)} g^{den}_{c\to n}, \quad R^{tot}_n = 1/g^{tot}_n$$

$$V_n = R^{tot}_n[g^L_n E^L + \Sigma_{i\in syn(n)} g_i x_i E^{rev}_i + \Sigma_{c\in child(n)} g^{den}_{c\to n} a_c], \quad a_c = f_c(V_c)$$

additive current

Changes the numerator without changing total conductance.

shunting conductance

Raises g^{tot}_n and lowers R^{tot}_n, even when net shunt current is small.

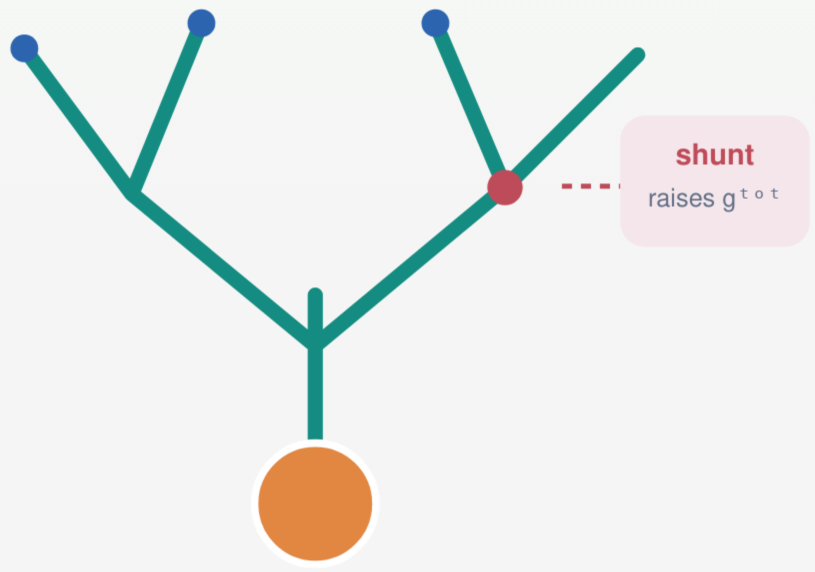
$V_n \approx E_I \Rightarrow g_I x_I(E_I - V_n) \approx 0$, but $g_I x_I$ still changes the denominator.

→ **Shunting changes local sensitivity because conductance appears in the voltage denominator.**

Holt & Koch (1997); Chance et al. (2002); Gidon & Segev (2012)

THE EXACT DENDRITIC GRADIENT

Dendritic credit still factorizes into eligibility × learning signal



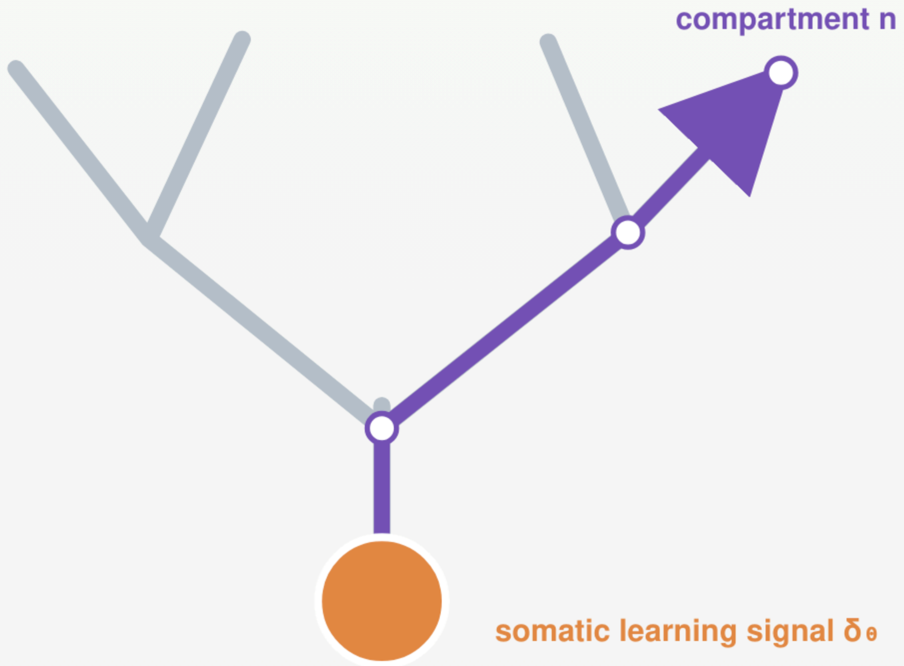
$\delta^V_n \equiv \partial\mathcal{L}/\partial V_n$ is the compartment learning signal

$$\partial V_n/\partial g_i = x_i R^{tot}_n (E^{rev}_i - V_n)$$

$$\partial\mathcal{L}/\partial g_i = \underbrace{x_i R^{tot}_n (E^{rev}_i - V_n)}_{\text{local dendritic eligibility } e^{den}_i} \times \underbrace{\delta^V_n}_{\text{returned compartment signal}}$$

→ **The local factor changes with dendritic state; the circuit-level learning signal still has to reach the correct compartment.**

Tree transport turns one somatic signal into a compartment field



$$\delta^V_0 = f'_0(V_0)\delta_u, \quad \delta^V_n = \delta^V_0 \gamma_n$$

$$\gamma_n = \prod_{(j\to k)\in \mathrm{path}(n\to 0)} f'_j(V_j)\, R^{tot}_k\, g^{den}_{j\to k}$$

directed tree

The product is indexed child→parent (n→0); returned credit propagates over the same path in reverse (0→n).

reconstructed reciprocal cable

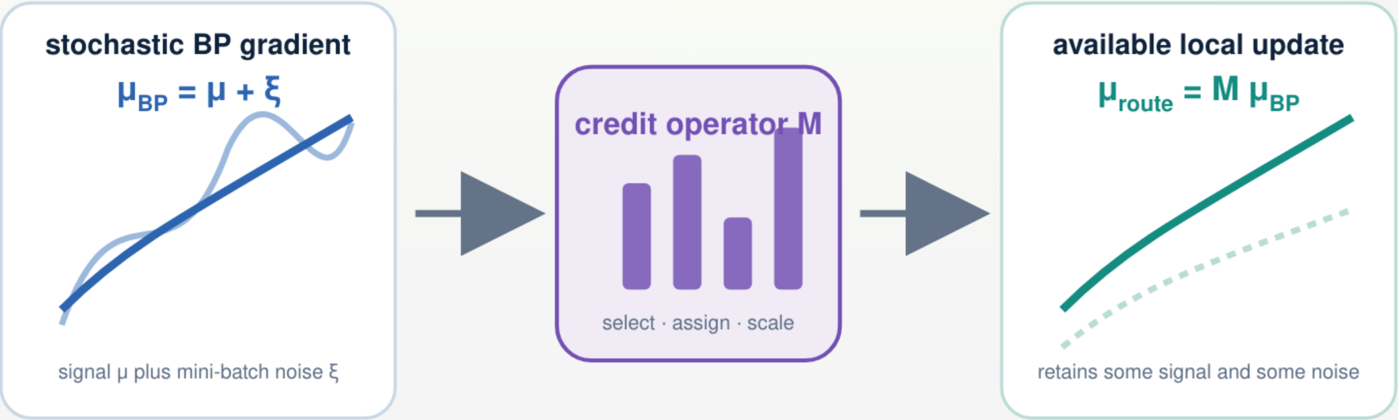
The corresponding field is obtained from the steady-state adjoint $J_V^T q = \nabla_V \mathscr{L}$.

→ **Dendrites do not generate the task error; they transform a neuron-level signal into spatially structured compartment credit.**

Almeida (1987); Pineda (1987); Schiess, Urbanczik & Senn (2016)

THE CREDIT OPERATOR

A feedback pathway selects, assigns, and scales stochastic credit



$$\mu BP=\mu+\xi,\quad [\xi]=0,\quad \mathrm{Cov}(\xi)=\Sigma$$

$$\mu route=M\mu BP,\quad w+=w-\eta\mu route$$

M = I: unrestricted backpropagation of the stochastic gradient. Restricted routes change the span, assignment, or gain of the available update.

→ **The operator M lets one theory describe scalar, neuron-specific, subtree-targeted, and exact compartment feedback.**

Operator utility balances retained signal, update cost, and noise

OPERATOR UTILITY · REQUIRES POSITIVE TASK ALIGNMENT

$$U(M)=\frac{[\mu^T M\mu]^2}{2L_{sm}[\|M\mu\|^2+\mathrm{tr}(M\Sigma M^T)]}$$

task-aligned signal · finite-step update cost · admitted stochastic noise

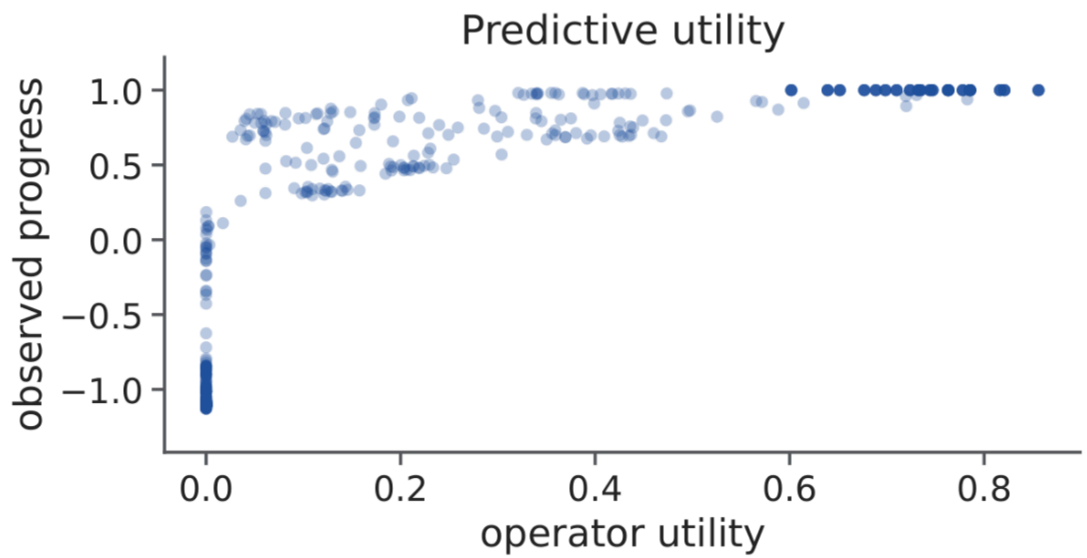
$\rho_S = 0.937$

utility versus observed norm-matched one-step progress across 540 trained conditions

$\rho_S = 0.916$

utility versus final trained accuracy

F



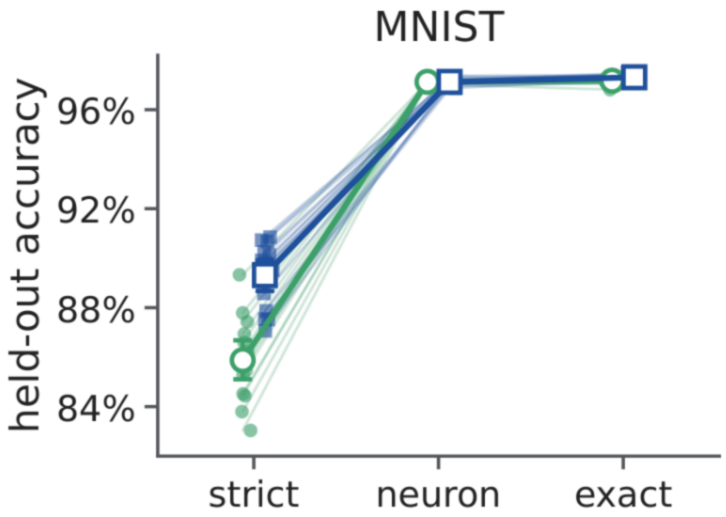
Exact for an isotropic quadratic with Hessian $L_{sm}I$; otherwise a curvature bound and one-step smoothness guarantee. It predicts large contrasts and within-family progress—not every trajectory-accrued difference.

→ **Restricted routing helps only when its retained, aligned signal compensates for the signal it discards and the noise it admits.**

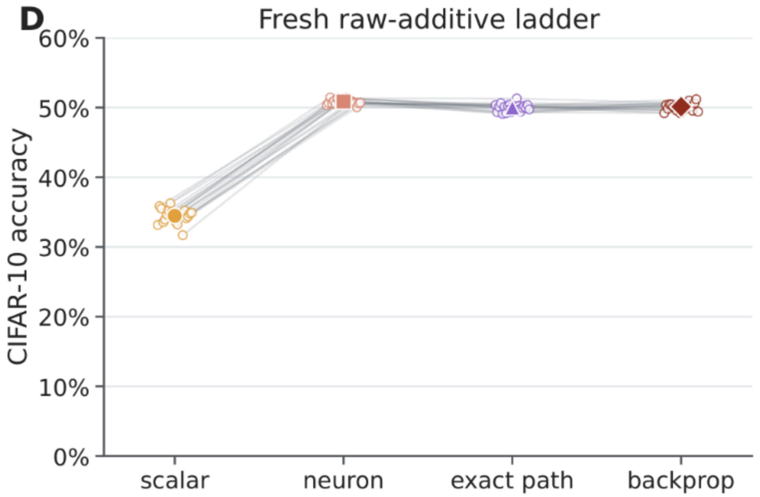
STANDARD IMAGE TASKS

Neuron identity—not exact path resolution—dominates both standard tasks

MNIST · strict scalar → neuron signal → exact compartment field



Flattened CIFAR-10 · additive tree · “exact path” = exact compartment field



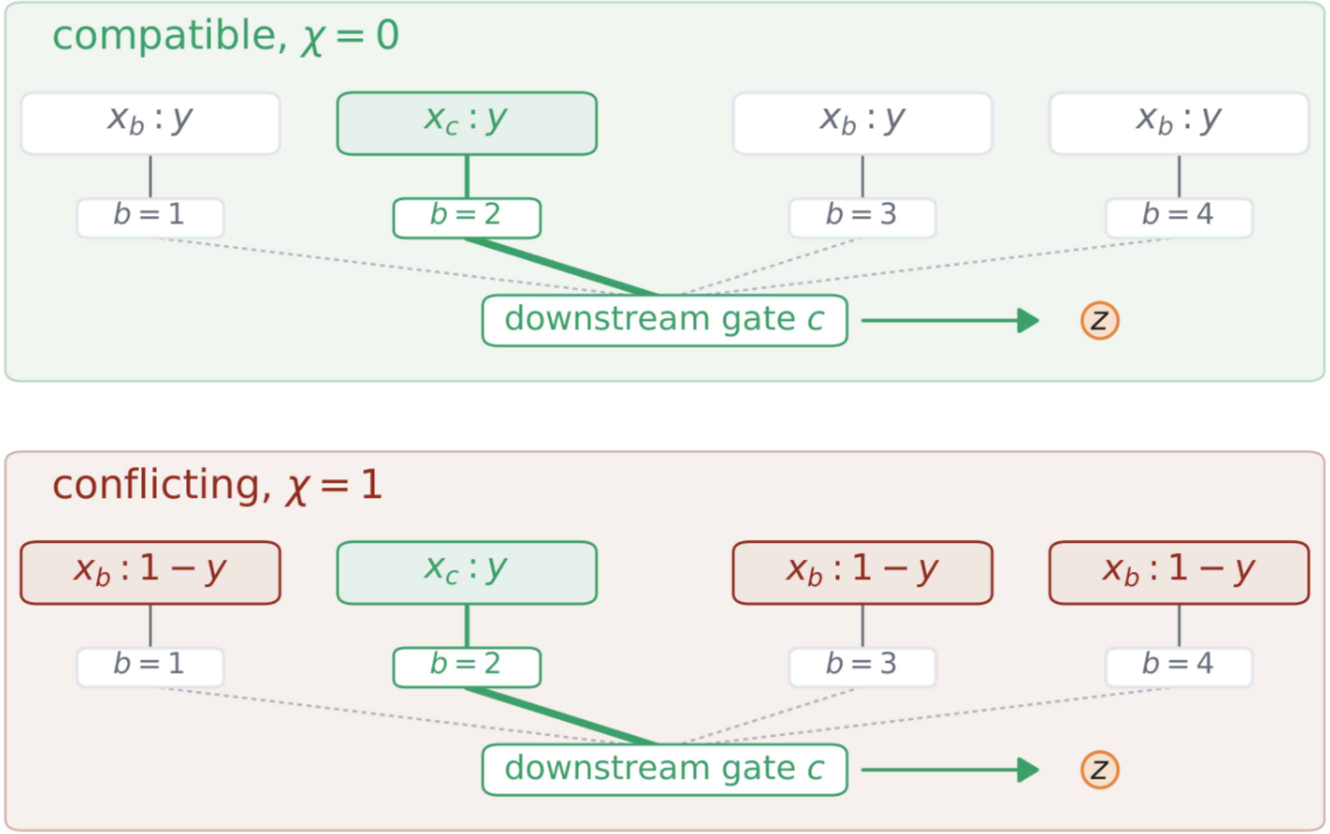
model/task	scalar → neuron	neuron → exact field
MNIST · shunting	+11.25 pp	**+0.045 pp**
MNIST · additive	+7.81 pp	**+0.186 pp**
CIFAR-10 · additive	+16.39 pp	–0.86 pp

→ **On both tasks, most of the gain comes from selecting the correct neuron; exact within-tree resolution adds little.**

Paired-seed means; CIFAR exact field and BP are equivalent within the predefined ±1-point margin.

CONTROLLED TASK 1 · DEFINITION

Credit conflict creates a demand for branch-specific signals



input

All B branches receive a Fashion-MNIST image and form nonzero eligibility.

forward selector

Context c selects the branch whose image determines the target.

conflict dose χ

Nonselected images vary from class-compatible (χ=0) to opposite-class (χ=1).

$$d^{\mathrm{branch}}{}_b = N^{-1}\Sigma_t\delta_t\,[c_t=b]x_{t,b}$$

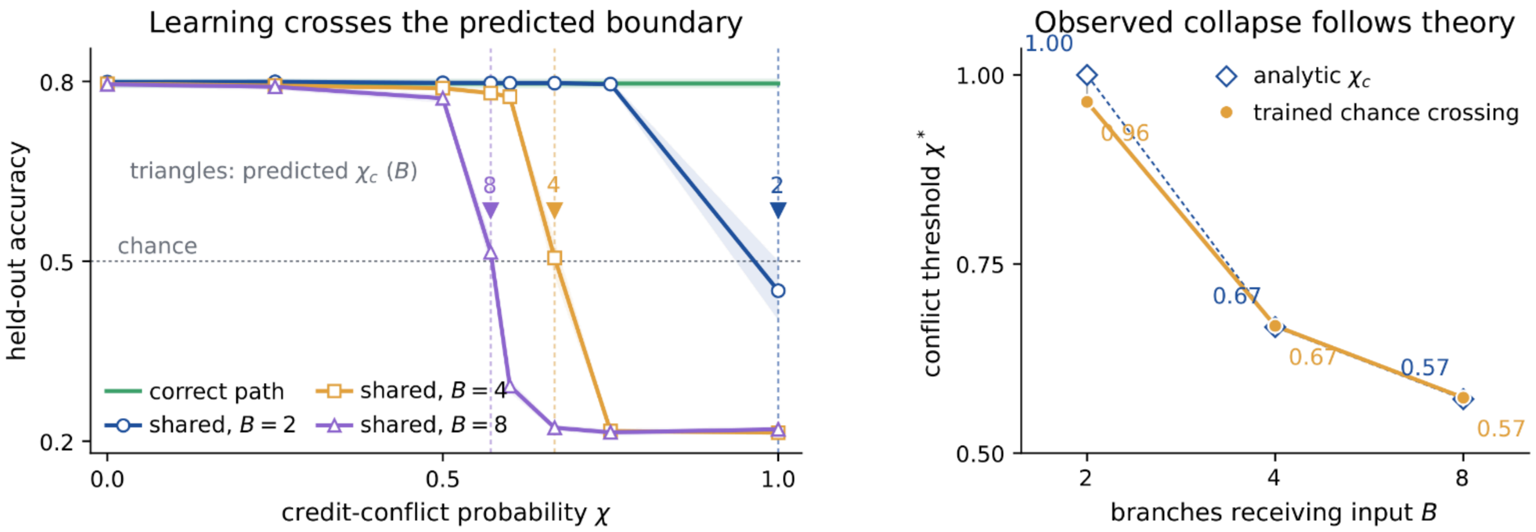
$$d^{\mathrm{shared}}{}_b = N^{-1}\Sigma_t\delta_t B^{-1}x_{t,b}$$

N: trials · **δ_t**: downstream logit gradient · **d_b**: mean update direction for branch b

At zero conflict, one shared signal is sufficient; at high conflict, different branches require opposite updates.

CONTROLLED TASK 1 · RESULT

Shared credit fails at the predicted conflict boundary



$\lambda_{shared} = B - 2\chi(B-1) \rightarrow \chi_c = B/[2(B-1)]$

35–58 pp

branch-specific gain over shared feedback at full conflict

causal control

Cyclically deranged routes fail despite identical rank and sparsity.

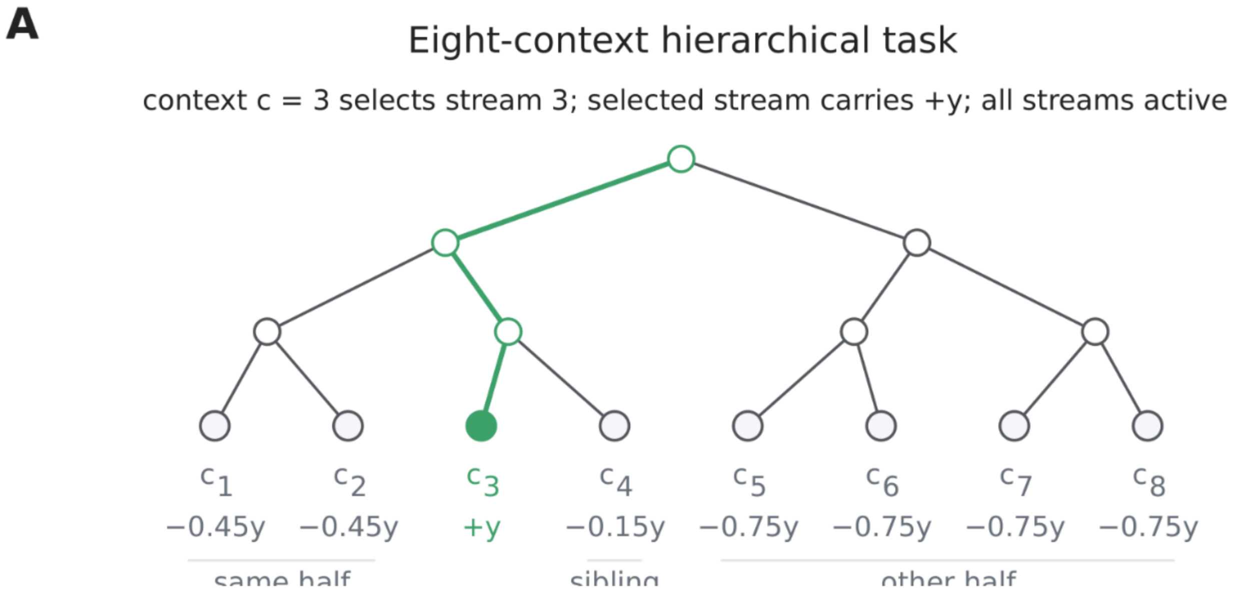
implementation control

Analytic BP, the correct route, and a gated-point calculation are equivalent forms of the same routing matrix.

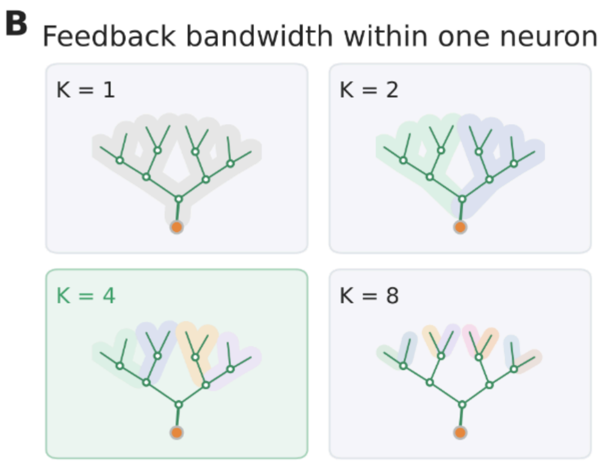
→ **The task establishes a need for an address—not a unique need for dendritic material.**

B = 2, 4, 8 branches; all empirical collapse points track the analytic boundary.

Nested tasks ask whether a few subtree addresses are efficient



$$\delta^{V,\,\mathrm{avail}} = A_K\beta, \quad \mathrm{rank}(A_K)=K$$

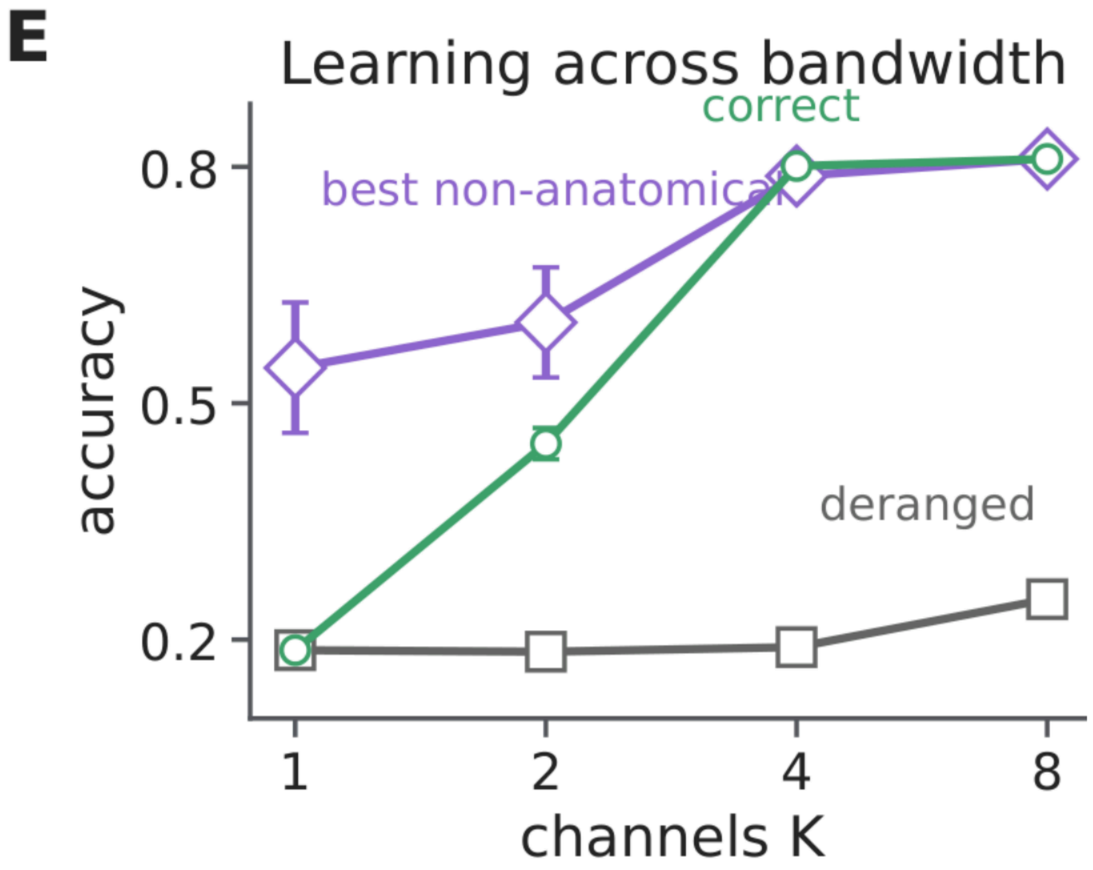


matched controls

Rank, sparsity, parameter count, and forward resources are held fixed; only route assignment, basis, or topology changes.

→ **Intermediate K tests whether tree-structured feedback is a useful low-dimensional basis for task credit.**

Subtree addresses help—but fine topology adds only a narrow gain



+61.1 pp
correct ancestry assignment over cyclic derangement at the same K=4 bandwidth

+1.27 pp
correct ancestry over the strongest matched non-anatomical low-rank control at K=4

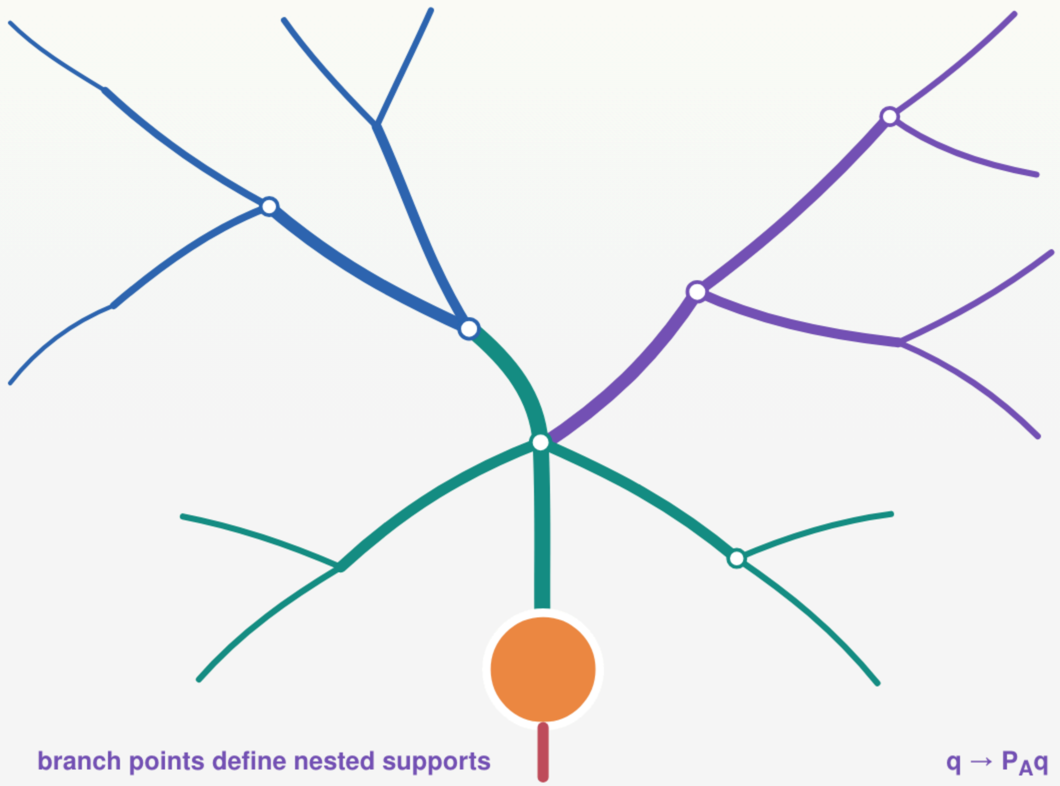
against the matched low-rank basis
Ancestry loses at K=1,2; wins only at K=4; and ties at full rank K=8. Rewiring removes the intermediate advantage.

equivalence
Dendritic, grouped-point, and gated-point implementations coincide for the same routed field.

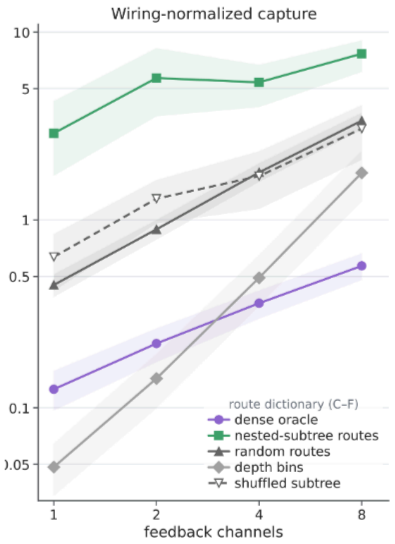
→ **The large effect is the value of within-neuron addresses; the anatomy-specific effect is conditional and modest.**

Twenty paired seeds; K=4 matched-topology contrast survives BH correction.

Real arbors provide sparse candidate routes, mostly through coarse geometry



$C_A(q)=\|P_A q\|^2/\|q\|^2$ fraction of field energy in the route span



85%
of dense field capture

≈ 7%
of dense route-matrix connections

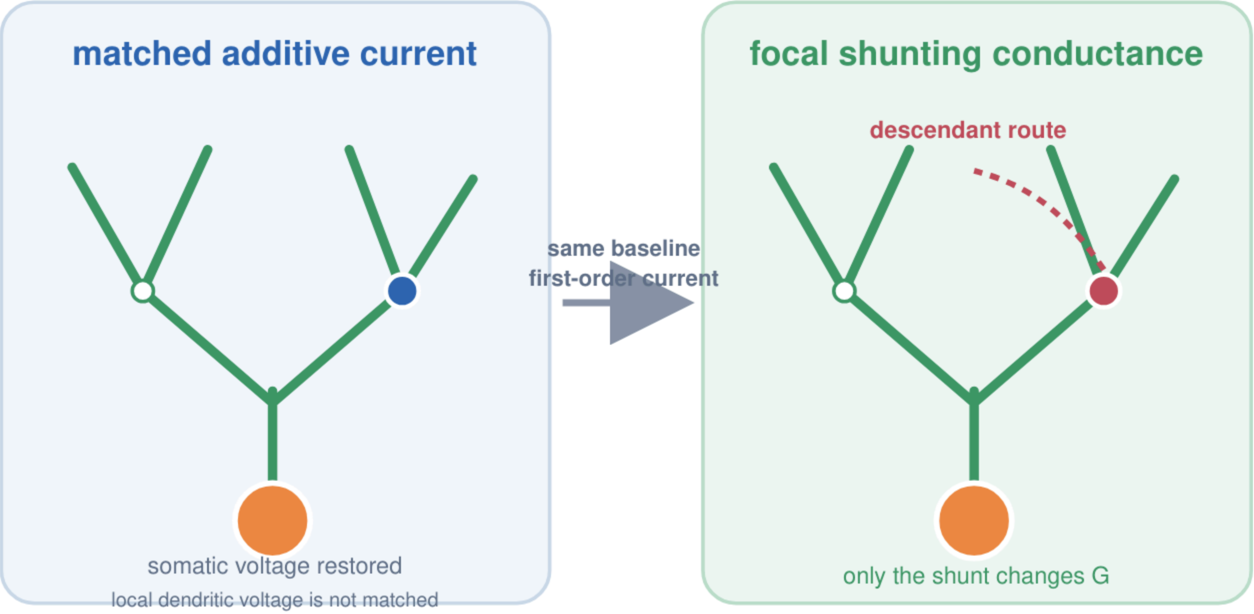
14.2×
dense capture per connection

≈ 2.7×
over a density-matched shuffled dictionary

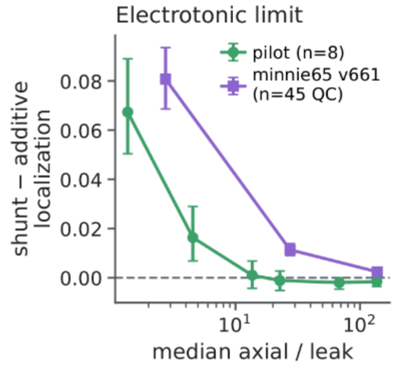
Connections are nonzero route-matrix entries—not cable length, energy, or reliability. Replicated in a disjoint 47-cell cohort and ten cells from a second MICrONS mouse; fields are modeled, not observed task gradients.

→ **Morphology supplies a sparse route dictionary; most capacity comes from branch depth and coarse topology.**

Focal shunting changes descendant credit only in permissive regimes



$$q' = q - [\kappa_k q_k/(1+\kappa_k(G^{-1})_{kk})]G^{-1}e_k$$



standard passive calibration

At R_m=15,000 Ω cm², the shunt-minus-current localization contrast is effectively zero.

high-conductance regime

Descendant-localized changes emerge and survive active-channel extensions.

→ **Shunting is a state-dependent route-gain mechanism, not a generic learning advantage.**

Baseline first-order focal current is matched and somatic voltage restored; local dendritic voltage is not matched.

MEASURED FUNCTIONAL EVIDENCE

Measured visual responses show no morphology-specific alignment

measured presynaptic responses

→

mapped conductances on each reconstructed target arbor

→

held-out postsynaptic-response prediction

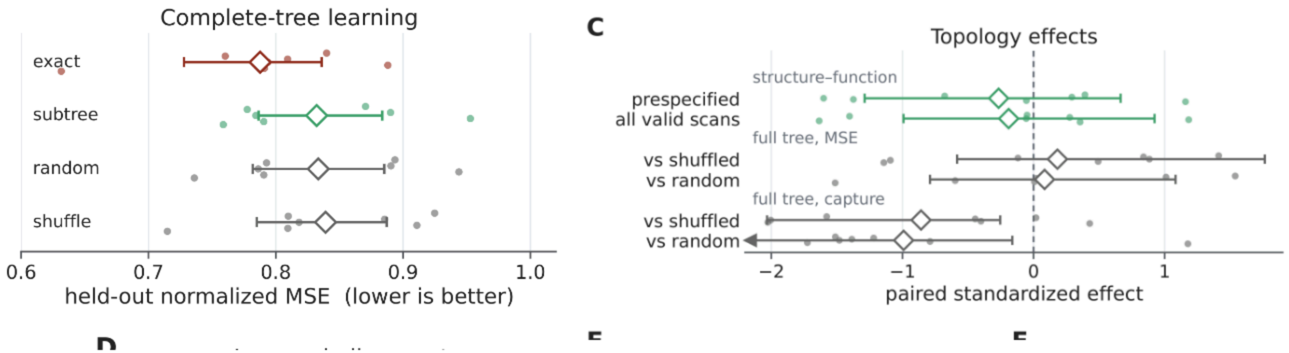
question

Does the measured input–output relation align more strongly with nested subtrees than with matched alternative route dictionaries?

inferential scope

Seven target cells from one MICrONS mouse; exact compartment errors are the reference.

Held-out learning and topology-minus-control effects



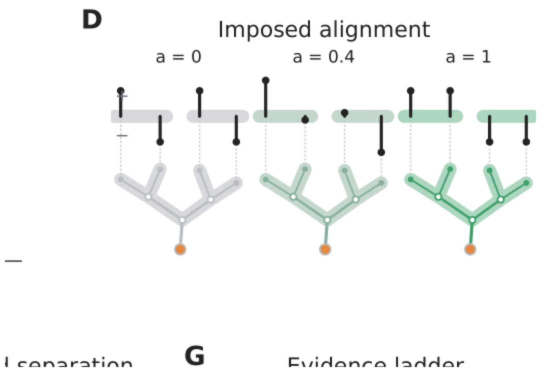
Nested subtrees do not outperform random or site-shuffled routes for held-out prediction, field capture, or within-arbor structure–function similarity.

→ **Measured anatomy supplies candidate routes, but these visual responses provide no evidence that the task uses them preferentially.**

The same anatomical routes capture task credit aligned to their span

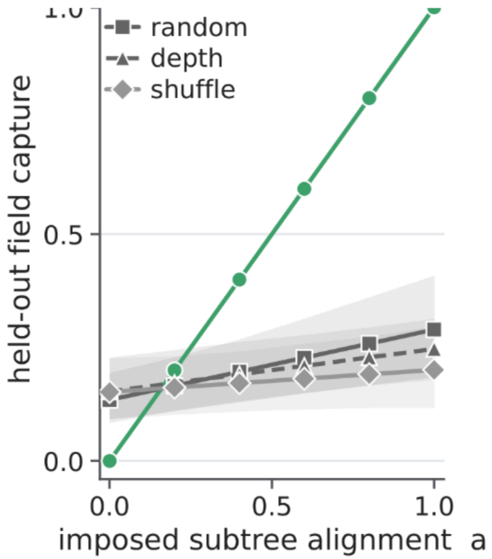
$$\varphi(a_{route}) = \sqrt{a_{route}}\, u_{\parallel} + \sqrt{(1-a_{route})}\, u_{\perp}$$

$u_{\parallel} \in \mathrm{col}(A),\quad u_{\perp} \perp \mathrm{col}(A),\quad \|u_{\parallel}\|=\|u_{\perp}\|=1$



controlled quantity

Anatomy, field energy, curvature, and route count are fixed; only alignment with the subtree span changes.



$$C_A[\varphi(a_{route})]=a_{route} \quad \text{by construction}$$

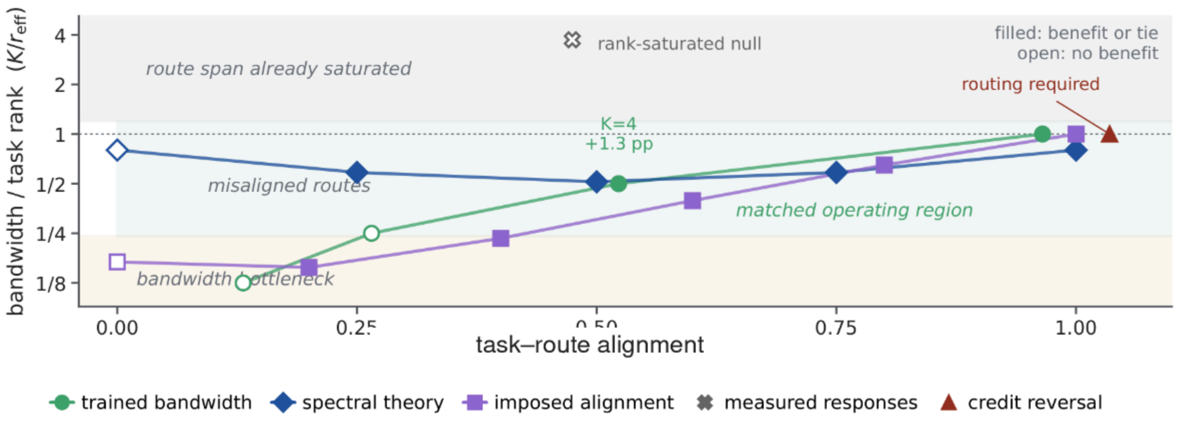
n = 8 reconstructed cells

Controls test whether the gain is specific to the true subtree span. The manipulation proves conditional representational sufficiency—not trained learning or endogenous biological use.

→ **Alignment is sufficient for representational capture in the model; endogenous morphology-specific alignment remains unestablished.**

SYNTHESIS

Alignment and bandwidth determine which dendritic resources help



relative bandwidth = K/r_{eff}, $r_{eff}=(\Sigma_i \lambda_i)^2/\Sigma_i \lambda_i^2$ · λ_i: task-credit-spectrum eigenvalues

Coordinates are estimated separately within each experiment. Regime tint and the $K/r_{eff}=1$ boundary are theoretical, not fitted.

neuron identity — SUPPORTED IN MODELS
dominant in standard tasks; retrospective animal contrast is consistent

subtree addresses — CONDITIONAL
large when local updates conflict; topology adds a narrow gain

route gain by shunting — CONDITIONAL
emerges only in permissive high-conductance regimes

anatomical route capacity — STRUCTURALLY SUPPORTED
sparse candidate routes on reconstructed arbors

endogenous morphology alignment — NOT ESTABLISHED
measured visual-response analyses are null

Neuron identity usually matters first; **subtree address** helps under task-aligned conflict; **shunting** changes route gain only in permissive electrotonic states.

→ **Dendritic structure is a conditional substrate for routing local credit—not a general replacement for backpropagation.**